

Soil 16S Tree-Reading Printout

Name: _____ Date: _____

Big idea: a phylogenetic tree is read by ancestry. To decide which tips are most closely related, find the most recent common ancestor, not the tips that look closest on the page.

Tips in this activity

Soil_ASV_A	Bacillus_subtilis_168
Soil_ASV_B	Rhizobium_leguminosarum_IAM12609
Pseudomonas_fluorescens_CCM2115	Streptomyces_coelicolor_rrnD
Acidobacterium_capsulatum_ATCC51196	

Simplified Teaching Tree

```
root
|
+-- Acidobacterium_capsulatum_ATCC51196
|
+-- Streptomyces_coelicolor_rrnD
|
+-- Soil_ASV_A
|   |
|   +-- Bacillus_subtilis_168
|
+-- Pseudomonas_fluorescens_CCM2115
|
+-- Soil_ASV_B
|   |
|   +-- Rhizobium_leguminosarum_IAM12609
```

For the full tree, use the notebook plots. This simplified drawing is for relationship-reading practice.

Vocabulary

Tip	A sequence or organism shown at the end of a branch.
Node	A branching point.
MRCA	Most recent common ancestor.
Sister taxa	Two tips or groups that share an immediate common ancestor.
Clade	An ancestor and all of its descendants.
Branch length	Sequence-distance information when drawn as a phylogram.

Questions

1. Which cached reference is Soil_ASV_A closest to?

Answer: _____

2. Which cached reference is Soil_ASV_B closest to?

Answer: _____

3. Is *Bacillus_subtilis_168* the ancestor of Soil_ASV_A, or are both tips descendants of a shared ancestor?

Answer: _____

4. Which idea is used to decide relatedness on a tree? A. page closeness B. color C. most recent common ancestor D. familiarity

Answer: _____

5. If a branch is rotated around a node, does the evolutionary relationship change? Explain.

Answer: _____

6. In this teaching tree, name one sister pair.

Answer: _____

7. Does a high 16S percent identity prove exact species identity? Explain.

Answer: _____

8. What does branch length show in this notebook's tree? A. p-value B. sequence distance C. abundance D. moisture

Answer: _____

9. What statistic is more appropriate for support on a tree branch? A. adjusted p-value B. bootstrap support C. read count D. sample depth

Answer: _____

10. Write one careful claim from this tree using: Soil_ASV_ ____ is closest to _____ in this cached 16S marker comparison, but this does not prove exact species identity.

Answer: _____

Exit Ticket

In one sentence, explain the difference between identifying a closest reference and showing evolutionary relatedness.

Instructor Answer Key

1. Soil_ASV_A is closest to *Bacillus_subtilis_168*.
2. Soil_ASV_B is closest to *Rhizobium_leguminosarum_IAM12609*.
3. Both are tips descended from a shared ancestor. The living reference is not the ancestor of the ASV.
4. C. Relatedness is decided by the most recent common ancestor.
5. No. Rotation around a node changes the drawing, not the relationship.
6. Accept Soil_ASV_A with *Bacillus_subtilis_168*, or Soil_ASV_B with *Rhizobium_leguminosarum_IAM12609*.
7. No. High 16S identity supports a closest-reference claim, but one short 16S region does not prove exact species identity.
8. B. Branch length is sequence-distance-derived branch length.
9. B. Bootstrap support is more appropriate for branch support. Adjusted p-values belong to abundance or metadata association tests.
10. Accept cautious claims such as: Soil_ASV_A is closest to *Bacillus_subtilis_168* in this cached 16S marker comparison, but this does not prove exact species identity.

Instructor Notes

- This printout is original and uses the soil 16S cache built for the Colab.
- It is inspired by tree-thinking skill targets from Baum, Smith, and Donovan, but does not reproduce their quiz pages.
- Emphasize: do not read trees as ladders of progress; do not decide relatedness by left-right order; do not treat tips as ancestors; do not call branch length a p-value.
- Use this before the Colab, then ask students to compare the simplified tree with the notebook's UPGMA and neighbor-joining plots.